

Literature Status for arXiv:0404401 Final Remark (A)

FA/Banach run packet

June 16, 2026

Status

Result type: literature already answered.

Source question: Piotr W. Nowak, *Coarse embeddings of metric spaces into Banach spaces*, arXiv:math/0404401, final remark (A).

Supporting answer: Manor Mendel and Assaf Naor, *Metric cotype*, arXiv:math/0506201.

Source Question

Nowak proves that $L_p(\mu)$ admits a coarse embedding into Hilbert space for $0 < p < 2$. In final remark (A), after recalling the Aharoni–Maurey–Mityagin uniform non-embedding theorem for L_p , $p > 2$, he asks whether the same phenomenon has a coarse-uniform analogue:

Does $L_p(\mu)$ admit a coarse uniform embedding into Hilbert space if $p > 2$?

Supporting Literature

Mendel–Naor introduce metric cotype and prove that, for Banach spaces, it coincides with classical Rademacher cotype. Their Theorem 1.6 states that if X has nontrivial type and Y coarsely embeds into X , then $q_Y \leq q_X$. In particular,

$$L_p \text{ embeds coarsely into } L_q \iff p \leq q \text{ or } q \leq p \leq 2.$$

The later corollary in the same paper gives the corresponding statement for both uniform and coarse embeddings.

Taking $q = 2$ gives a negative answer for $p > 2$: the conditions $p \leq 2$ and $2 \leq p \leq 2$ both fail. Thus L_p does not coarsely embed into Hilbert space for $p > 2$. Since a coarse-uniform embedding is stronger than a coarse embedding, Nowak’s coarse-uniform question has a negative answer in this standard infinite-dimensional L_p setting.

Formal Statement

Theorem. *Final remark (A) of Nowak, arXiv:math/0404401, is answered negatively in the later literature for the standard infinite-dimensional L_p spaces. For $p > 2$, L_p does not admit a coarse embedding, hence does not admit a coarse-uniform embedding, into a Hilbert space.*

Proof. Apply Mendel–Naor’s coarse embedding classification with $q = 2$. For $p > 2$, neither $p \leq 2$ nor $2 \leq p \leq 2$ holds, so their theorem rules out a coarse embedding $L_p \rightarrow L_2$. A Hilbert target is an L_2 -space up to linear isometry. Therefore L_p , $p > 2$, does not coarsely embed into Hilbert space. Any coarse-uniform embedding would in particular be a coarse embedding, so it is also impossible. $\square$

Scope and Caveat

The source writes $L_p(\mu)$. The supporting theorem is stated for the standard L_p -scale and covers the intended infinite-dimensional examples, including $L_p[0, 1]$ and ℓ_p . If $L_p(\mu)$ is read literally to include finite-dimensional measure spaces, those finite-dimensional cases embed into Hilbert space after changing constants and are a trivial exception outside the intended question.

This packet records a known literature answer rather than a new proof. The supporting paper explicitly notes that Johnson–Randrianarivony had already resolved the $p > 2$ Hilbert case negatively and then proves the stronger classification above.

Verification

Stored files:

`source_paper.pdf`, `supporting_paper_0506201.pdf`.

The source question appears in the parsed TeX

`data/parsed/arxiv_sources/0404401/source.tex`

around final remark (A). The supporting classification appears in

`data/parsed/arxiv_sources/0506201/source.tex`

as Theorem `thm:coarse` and the later corollary proving that L_p embeds uniformly or coarsely into L_q iff $p \leq q$ or $q \leq p \leq 2$.

Cheap-index search before packaging found no prior exact packet for arXiv:0404401 or this L_p , $p > 2$, Hilbert coarse-embedding question.

References

- [1] P. W. Nowak, *Coarse embeddings of metric spaces into Banach spaces*, arXiv:math/0404401. <https://arxiv.org/abs/math/0404401>
- [2] M. Mendel and A. Naor, *Metric cotype*, arXiv:math/0506201. <https://arxiv.org/abs/math/0506201>
- [3] W. B. Johnson and N. L. Randrianarivony, ℓ_p ($p > 2$) does not coarsely embed into a Hilbert space, Proc. Amer. Math. Soc. 134 (2006), no. 4, 1045–1050.